Table of Contents

Temperature controller calculations 1
Settings 1
Feedforward fraction 1
PI trim 1
Hot duty fraction 2
Valve switching 2
Unequal flows (explanation only, not in the controller) 3

Temperature controller calculations

Ruben Haasbroek, 26865238. Controller from pid.py.

```
clear; clc;
```

Settings

```
Kp = 0.15;      % 1/degC
Ki = 0.015;     % 1/(degC s)
P  = 6;         % s
dt = 0.5;       % s
umin = -1;
umax = 1;
```

Feedforward fraction

$$f_{ff} = \frac{T_{set} - T_{cold}}{T_{hot} - T_{cold}} \qquad (D.4.1)$$

```
Tset = 37;  Thot = 45.9;  Tcold = 25.3;
fff = (Tset - Tcold)/(Thot - Tcold);
fprintf('f_ff = (%.1f - %.1f)/(%.1f - %.1f) = %.3f\n', Tset, Tcold, Thot,
Tcold, fff);
```

```
f_ff = (37.0 - 25.3)/(45.9 - 25.3) = 0.568
```

PI trim

$$e = T_{set} - T_{mix} \qquad (D.4.2)$$

$$I^* = I + e\,\Delta t \qquad (D.4.3)$$

$$u = K_p\,e + K_i\,I^* \qquad (D.4.4)$$

The integral only updates when u is between -1 and 1 (D.4.5). The trim is then limited to -1 and 1 (D.4.6).

```
% Example 1: start-up, outlet at 28.0 degC, integral at zero
Tmix = 28.0;  I = 0;
```

```
e = Tset - Tmix;
Istar = I + e*dt;
u = Kp*e + Ki*Istar;
fprintf('e = %.1f degC, I* = %.2f, u = %.4f\n', e, Istar, u);
if u >= umin && u <= umax
    I = Istar;
else
    fprintf('u is outside the limits, integral stays at %.1f\n', I);
end
trim1 = min(umax, max(umin, u));
d1 = min(1, max(0, fff + trim1));
fprintf('trim = %.1f, d = %.3f\n', trim1, d1);

% Example 2: outlet at 39.0 degC, integral at 10 degC s
Tmix = 39.0;  I = 10;
e = Tset - Tmix;
Istar = I + e*dt;
u = Kp*e + Ki*Istar;
fprintf('e = %.1f degC, I* = %.2f, Kp*e = %.3f, Ki*I* = %.3f, u = %.3f\n', e,
Istar, Kp*e, Ki*Istar, u);
```

```
e = 9.0 degC, I* = 4.50, u = 1.4175
u is outside the limits, integral stays at 0.0
trim = 1.0, d = 1.000
e = -2.0 degC, I* = 9.00, Kp*e = -0.300, Ki*I* = 0.135, u = -0.165
```

Hot duty fraction

$d = f_{ff} + u$ (D.4.7) with d limited to 0 and 1.

```
d2 = min(1, max(0, fff + u));
fprintf('d = %.3f + (%.3f) = %.3f\n', fff, u, d2);
fprintf('hot time in one period = %.2f s of %g s\n', d2*P, P);
```

```
d = 0.568 + (-0.165) = 0.403
hot time in one period = 2.42 s of 6 s
```

Valve switching

The hot valve is open while

$\tau < d\,P$ (D.4.8)

where tau is the time into the 6 s period. Example where d changes during the period:

```
tau = 0:dt:P-dt;
d = 0.50*ones(size(tau));
d(tau >= 2.5) = 0.35;
d(tau >= 3.5) = 0.70;
hot = tau < d*P;
fprintf('tau   :'); fprintf(' %4.1f', tau); fprintf('\n');
fprintf('d     :'); fprintf(' %4.2f', d);   fprintf('\n');
fprintf('hot   :'); fprintf(' %4d', hot);   fprintf('\n');
fprintf('valve changes in the period: %d\n', sum(diff(hot) ~= 0));
```

```
tau   :  0.0  0.5  1.0  1.5  2.0  2.5  3.0  3.5  4.0  4.5  5.0  5.5
d     : 0.50 0.50 0.50 0.50 0.50 0.35 0.35 0.70 0.70 0.70 0.70 0.70
hot   :    1    1    1    1    1    0    0    1    1    0    0    0
valve changes in the period: 3
```

Unequal flows (explanation only, not in the controller)

$$T_{mix} = \frac{d\,Q_h\,T_{hot} + (1-d)\,Q_c\,T_{cold}}{d\,Q_h + (1-d)\,Q_c} \qquad (D.4.9)$$

```
Qh = 4.557;  Qc = 5.180;    % L/min
Tmix = (fff*Qh*Thot + (1-fff)*Qc*Tcold)/(fff*Qh + (1-fff)*Qc);
fprintf('T_mix = %.2f degC, %.2f degC below the setpoint\n', Tmix, Tset -
Tmix);
```

```
T_mix = 36.35 degC, 0.65 degC below the setpoint
```

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